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Terranes or Cambrian polar wander: New data from the Scott Glacier area, Transantarctic Mountains, Antarctica

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Abstract. We present new paleomagnetic and isotopic data from the Transantarctic Mountains in East Antarctica that constrain the paleogeographic position of this region during the late Early and Middle Cambrian. Two new poles have been determined from volcanic and granitic rocks in the Scott Glacier area. The first pole is from the Wyatt and Ackerman Formations (~525 Ma) and the Mount Paine tonalite (40°E, 1°N, A95= 6° (error circle at the 95% confidence level), N=11 VGPs). The second pole is from the Zanuck granite (36°E, 7°S, A95= 9°, N=9 VGPs). These poles differ from the Gondwana early Paleozoic reference poles and could indicate that the Scott Glacier area (and by geologic correlation, north Victoria Land, eastern Australia, and West Antarctica?) was part of a terrane accreted to East Gondwana in the Cambrian. Another possibility is that these new poles support rapid apparent polar wander in the late Early Cambrian and Middle Cambrian. This apparent polar wander event could be related to rapid plate motions or to a true polar wander event. Lastly, there may have been a local vertical axis rotation of the Scott Glacier area if the Wyatt pole is compared with the African Ntonya pole.

1. Introduction

We undertook a field mapping and sampling program in the Transantarctic Mountains of East Antarctica to better understand Antarctica's tectonic role during the final stages of Gondwana assembly in the Early Paleozoic. We hoped that this study would further constrain the early Paleozoic Gondwana apparent polar wander path [*Grunow*, 1995; *Meert and Van der Voo*, 1996] and establish the isotopic age of rocks thought to be neo-proterozoic. In addition, obtaining new Cambrian reference poles from Antarctica could further constrain Gondwana's position relative to North America and South America, hence testing models predicting Laurentia's connection to Gondwana [*Dalziel*, 1991; *Hoffman*, 1991].

Our sampling goals were to obtain paleomagnetic and geochronologic samples from neo-proterozoic? and Early Paleozoic igneous and sedimentary rocks from the Scott Glacier area (Figure 1). The U-Pb isotopic dating demonstrated that the igneous rocks previously thought to be Neo-proterozoic are actually Cambrian [*Encarnación and Grunow*, 1996]. In this paper we present new paleomagnetic data from late Lower Cambrian and Middle Cambrian rocks constituting the first Cambrian paleomagnetic results from the Transantarctic Mountains.

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2. Regional Setting

The geology of the Scott Glacier area (Figure 1) consists primarily of Neo-proterozoic? La Gorce Formation metagraywackes, late Lower Cambrian volcanic and clastic metasedimentary rocks of the Wyatt and Ackerman Formations, and Cambro-Ordovician granitoids. The greenschist facies, folded slates and metasandstones of the La Gorce Formation are overlain unconformably by the Wyatt and Ackerman Formations, respectively [*Stump*, 1995].

The Wyatt Formation consists of dacitic igneous rocks thought to have been emplaced as ignimbritic sheets during volcanic eruptions (caldera collapse) and as hypabyssal intrusions [*Borg*, 1980; *Stump et al.*, 1986]. Most of the Wyatt Formation is massive although some outcrops exhibit felsic lenses that are considered to be fiamme parallel to primary volcanic layering [*Borg*, 1980]. In thin section most of the Wyatt Formation contains phenocrysts of quartz + feldspar + biotite ± rock fragments in an aphanitic groundmass [*Borg*, 1980]. The Pb-Pb age on zircons from a Wyatt Formation ash flow tuff (site LGC5A on Ackerman Ridge) is 526 ± 2 Ma (2 σ error based on three concordant fractions) [*Encarnación and Grunow*, 1996].

Found structurally between the Wyatt and La Gorce Formations on Ackerman Ridge (Figure 1) is the Ackerman Formation. The Ackerman Formation appears to lie conformably above the Wyatt Formation although rubble obscures the contact. The Ackerman Formation represents a very localized occurrence of shallow marine sandstones, siltstones, red and green shales, lava flows, and volcaniclastic rocks formed in a volcanic arc environment [*Stump*, 1983; *Stump et al.*, 1986]. The age of the Ackerman Formation (from lava flow, LGC5C) has been determined to be 524 ± 2 Ma (2 σ, four concordant Pb-Pb zircon analyses) [*Encarnación and Grunow*, 1996]. The agreement between the Ackerman and Wyatt Formations' U-Pb ages argues for their nearly synchronous emplacement and supports the view that the contact is a conformable one. Both the Ackerman and Wyatt Formations have been variably altered either by original magmatic fluids within the volcanic complex or by regional, very low grade metamorphism [*Stump et al.*, 1986].

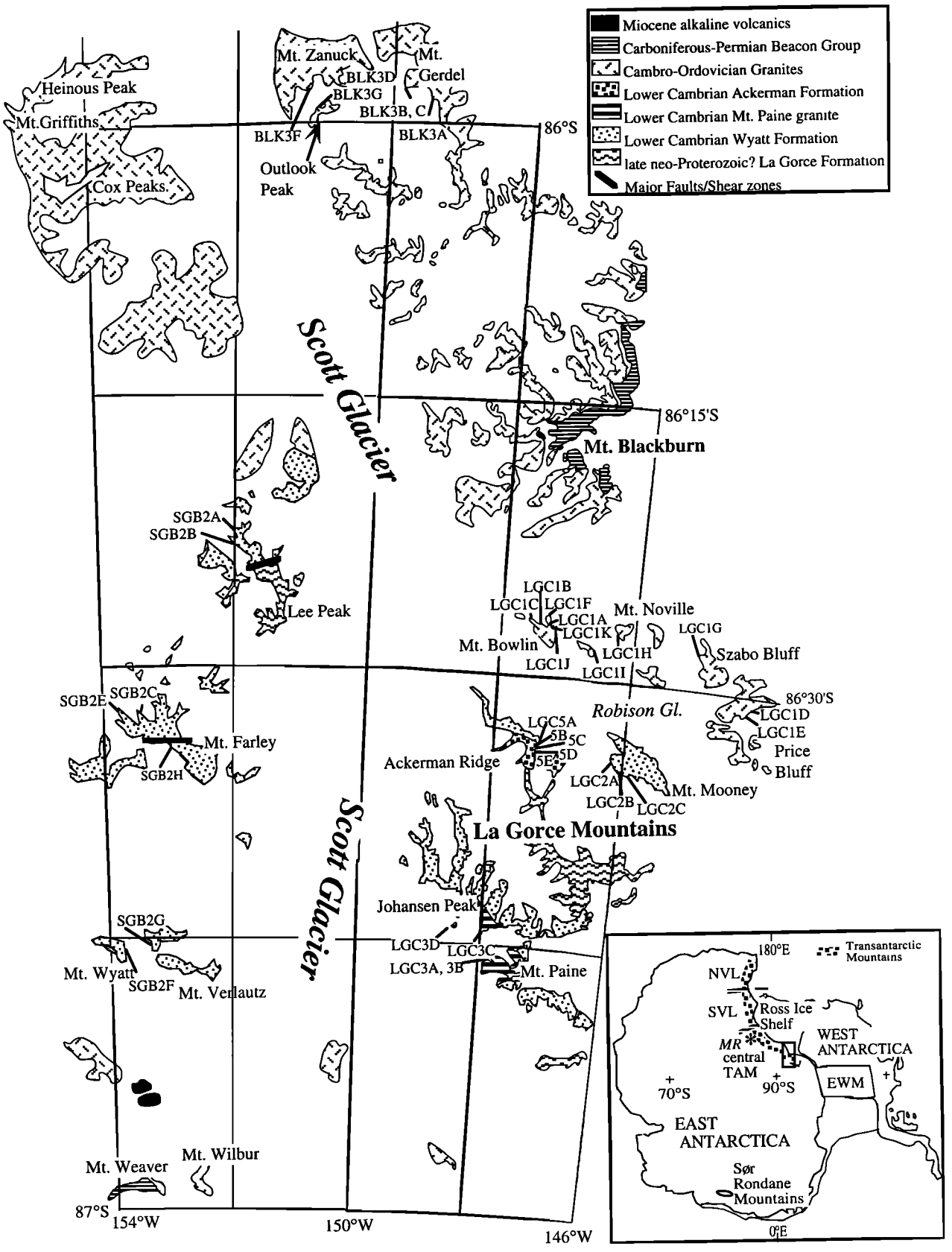


Figure 1. Paleomagnetic locations from the Scott Glacier region. Inset shows the location of paleomagnetic studies in the Scott Glacier area, the Sør Rondane Mountains, and an outline of the main Antarctic physiographic boundaries. NVL, north Victoria Land; SVL, south Victoria Land; EWM, Ellsworth-Whitmore Mountains block; MR, Miller Range; TAM, Transantarctic Mountains.

Structurally, the Wyatt Formation has been sheared in some locations (i.e., Mount Farley) whereas in other places, the rocks are massive and undeformed (i.e., Mount Wyatt). The narrow, vertical ductile shear zones observed locally in the Wyatt Formation generally exhibit down dip stretching lineations. At Ackerman Ridge the Ackerman Formation has been locally folded, cleaved, and faulted, but on the basis of sedimentary structures, the 2000 m Ackerman Formation section is largely upright. Folding in the Ackerman Formation sandstones and shales versus localized shear deformation observed in the Wyatt Formation may be due to the large competency contrast between the lithologies. The deformation in the Wyatt and Ackerman Formations is considered to be related to the Cambrian Ross Orogeny in the Transantarctic Mountains [*Encarnación and Grunow*, 1996].

The Wyatt Formation is intruded by calc-alkaline granitic plutons known as the Queen Maud Batholith [*Borg*, 1983]. The quartz-dioritic to granitic intrusions contain roof pendants of the Wyatt and Ackermann Formations [*Katz and Waterhouse*, 1970]. The plutons exhibit sharp, discordant contacts, stoped blocks, and narrow contact aureoles, suggesting emplacement at shallow levels in the crust [*Borg*, 1983].

The oldest and largest pluton in the region is the megacrystic Zanuck granite, which intrudes the Wyatt Formation at Heinous Peak [*Stump*, 1995] (Figure 1). Pressure and temperature estimates from andalusite-bearing pegmatites found within the Zanuck granite at Szabo Bluff suggest emplacement depths of 4 kbars at 650°C [*Burt and Stump*, 1983] (Figure 1). In general, the Zanuck granite displays a well-developed magmatic flow foliation, but there are localized zones of post magmatic ductile shearing and cataclasis. The Zanuck granite is in turn cut by younger, undeformed, equigranular, medium-grained, biotite granites. A younger granite at Mount Wilbur yields a K/Ar biotite age of 470 ± 14 Ma [*Minshew*, 1971]. More recent $^{40}Ar/^{39}Ar$ dating on biotites from granitoids in the Scott Glacier area yielded apparent ages ranging from 460 to 480 Ma [*Baldwin et al.*, 1995]. Most or all of the granites regionally appear to be related to early Paleozoic subduction beneath Antarctica [*Borg et.al.*, 1990]. The undeformed, equigranular granitoids in the Scott Glacier area indicate that deformation had probably ceased by the time of their emplacement in the Early to Middle Ordovician. In the central Transantarctic Mountains (TAM), the Ross Event deformation ceased by around 500 Ma [*Encarnación and Grunow*, 1996], consistent with the lack of deformation in the Ordovician granitoids.

The Cambro-Ordovician igneous and sedimentary/volcanic rocks are overlain unconformably by flat-lying Carboniferous rocks of the Beacon sequence [*Minshew*, 1967]. Sills of Early Jurassic Ferrar diabase [*Minshew*, 1967; *Encarnación et al.*, 1996] intrude the Beacon strata. Miocene alkaline volcanism has occurred in a very small, localized area at the head of the Scott Glacier [*Stump* et al., 1980] (Figure 1).

3. Paleomagnetic Sampling and Laboratory Procedures

Over 500 paleomagnetic cores were collected from lower Paleozoic igneous rocks within a 100 x 150 km^2 area of the Scott Glacier [*Grunow and Encarnación*, 1995]. Samples were taken from similar age intrusive and extrusive rocks over a large region to average out secular variation and evaluate the possibility of structural disruption. Usually six to seven cores were sampled at each site using a Pomeroy, gasoline-powered, rock-coring drill. The samples were oriented with a Brunton magnetic compass. Sun compass measurements were made at each locality to allow correction for local magnetic variation and were always within 0° - 3° of the predicted magnetic variation. Usually 4 to 8 sites were collected within each pluton, generally from sites several kilometers apart, as a way to evaluate tilting/rotation of the rocks. The geographic coordinates of some locations were determined from a handheld Global Positioning System (GPS) "Magellan" receiver and are listed to 0.01° in Table 1.

Analyses of specimens (cut from each core) were performed at the Byrd Polar Research Center paleomagnetism laboratory. A JR5A magnetometer was used to measure the natural remanent magnetization (NRM) on one or more specimens from each core. Pilot samples were progressively demagnetized using step wise alternating field (AF) or thermal (TH) techniques. All samples were then demagnetized using the best demagnetization procedure (either AF or TH) for each site. During AF demagnetization, steps of generally 5 - 10 mT up to a peak of 90 mT were used (minimum of eight steps), while during thermal demagnetization steps, of generally 50° or 100°C up to 500°C and then smaller intervals were used (minimum of 10 steps). Susceptibility measurements were made on a Bartington MS2 susceptiblity bridge after each thermal demagnetization step to evaluate mineralogical changes associated with heating the specimens. Isothermal remanent magnetization (IRM) experiments were performed on representative specimens from the main sample locations. Magnetic saturation was obtained in fields of less than 300 mT and when combined with the observed unblocking temperature spectra suggests that the sample remanence is carried by titanomagnetite in the igneous rocks.

The magnetization directions were determined by principal component analysis for collinear segments converging toward the origin [*Kirschvink*, 1980] plotted on vector end point diagrams. Magnetization directions were considered stable (whether primary or secondary) if defined by three or more collinear points on the vector component diagrams and produced maximum angular deviation values for linear segments of less than 15° (usually less than 10°). The mean characteristic magnetization directions for each locality were obtained by combining sample means to site means (Table 1) and then site means to a unit mean (Table 2) using standard *Fisher* [1953] statistics. Site mean directions with a95s ≥ 15° were not used for calculating unit mean directions or virtual geomagnetic poles (VGPs). The regional Scott Glacier Cambrian poles were calculated from the site mean VGP poles (Table 2).

4. Results

4.1. Wyatt Formation

The Wyatt Formation was sampled at Mount Mooney, Ackerman Ridge, Mount Wyatt, the Mount Verlautz area, Mount Farley, and the end of the ridge near Lee Peak (Figure 1). At Mount Verlautz and Mount Wyatt, field relations suggest that the massive Wyatt Formation is a hypabyssal intrusion whereas the Wyatt Formation near Lee Peak, Ackerman Ridge, and Mount Mooney is considered to be volcanic [*Borg*, 1980]. The 15° dip of the layering defined by felsic lenses near Lee

Table 1. New Scott Glacier Site Mean Directions and Virtual Geomagnetic Poles

	Site Coordinates			Tilt-			NRM x						Paleopole	
	Latitude, deg	Longitude, deg.	Site	Correction	Lithology	Component	10-2 A/m	*n/N*	*D* , deg	*I* , deg	a95	*k*	Longitude, deg.	Latitude, deg.
					La Gorce Mountains									
Mt. Bowlin	86°28'S	147°18' W	LGC1A		Zanuck granite		0.10	4\5	165.7°	8.0°	9.9°	87.2	18.3°	-7.4°
Mt. Bowlin	86°28'S	147°18' W	LGC1B		Zanuck granite		1.86	7\7	174.0°	2.3°	3.8°	248.0	26.7°	-4.6°
Mt. Bowlin	86°28'S	147°18' W	LGC1C		Zanuck granite		0.54	7\8	178.9°	7.8°	5.4°	124.1	31.6°	-7.8°
Mt. Bowlin	86°28'S	147°18' W	LGC1D		Zanuck granite		0.02	7	too weak	-	-	-	-	-
Price Bluff	86°32'S	144°34'W	LGC1E		Zanuck granite		0.05	3\6	351.5°	-8.0°	10.4°	141.4	206.9°	7.5°
Price Bluff	86°32'S	144°34'W	LGC1F		Zanuck granite		0.27	6\7	171.1°	13.2°	6.3°	115.2	26.4°	-10.1°
Szabo Bluff	86°29'S	144°34'W	LGC1G		Zanuck granite		0.14	6	unstable	-	-	-	-	-
Mt. Noville	86°27'S	146°10'W	LGC1H		mafic enclaves		0.10	7\8	189.0°	22.8°	7.6°	63.9	43.0°	-15.4°
Mt. Bowlin	86°27.96'S	146°33.08'W	LGC1I		Bowlin granite		2.93	7	unstable	-	-	-	-	-
Mt. Bowlin			LGC1J		Bowlin granite		1.84	7	unstable	-	-	-	-	-
Mt. Bowlin			LGC1K		Bowlin granite		1.51	7	unstable	-	-	-	-	-
Mt. Bowlin	86°28.34'S	146°42.42'W	LGC1L		Bowlin granite		3.68	7	unstable	-	-	-	-	-
Mt. Paine	86°46°S	147°32'W	LGC3A		quartz diorite		0.07	6\8	185.1°	-16.2°	13.4°	26.0	37.6°	5.1°
Mt. Paine	86°46°S	147°32'W	LGC3B		quartz diorite		0.05	6\7	181.2°	-6.5°	13.3°	26.0	33.7°	0.0°
Johansen Peak	86°44.15'S	148°01.39'W	LGC3C		intrusive Wyatt		0.10	7\7	189.1°	-13.8°	3.5°	297.0	41.6°	3.8°
Johansen Peak	86°44'S	148°01'W	LGC3D		intrusive Wyatt		0.08	7	unstable	-	-	-	-	-
Ackerman Ridge	86°35'S	148°20'W	LGC5A		Wyatt Fm.	A	0.77	6\6	189.8°	15.5°	14.3°	22.8	42.3°	-11.3°
Ackerman Ridge	86°35'S	148°20'W	LGC5A	151/36W*	Wyatt Fm.	A		6\6	191.7°*	-8.0°*	14.3°*	22.8	44.0°*	0.7°*
Ackerman Ridge	86°35'S	148°20'W	LGC5B		Ackerman Fm. Lava 1	A	0.30	6\7	180.1°	6.0°	10.6°	40.5	32.4°	-6.4°
Ackerman Ridge	86°35'S	148°20'W	LGC5B	151/36W*	Ackerman Fm. Lava 1	A		6\7	178.5°*	-11.5°*	10.6°*	40.6	30.8°*	2.4°*
Ackerman Ridge	86°35'S	148°20'W	LGC5C		Ackerman Fm. Lava 2	A	0.64	4\7	169.4°	45.9°	14.6°	40.8	21.4°	-30.7°
Ackerman Ridge	86°35'S	148°20'W	LGC5C	114/64W*	Ackerman Fm. Lava 2	A		4\7	180.2°*	-11.5°*	14.6°*	40.8	32.6°*	2.4°*
Ackerman Ridge	86°35'S	148°20'W	LGC5D		Ackerman Fm. Lava 4	A	0.78	7\8	194.3°	25.2°	11.2°	30.0	46.9°	-16.6°
Ackerman Ridge	86°35'S	148°20'W	LGC5D	140/50W*	Ackerman Fm. Lava 4	A		7\8	196.5°*	-16.8°*	11.2°*	30.1	48.8°*	5.3°*
			LGC5 combined			C		19	10.6°	-81.7°	10.3°	11.6	225.7°	77.1°
			LGC5 combined			C		19	41.5°	-39.2°	13.1°	7.5	254.9°	24.7°
					Southwestern Scott Glacier Region									
Lee Peak area	86°22.16'S	152°03.01'W	SGB2A		felsic Wyatt Fm.		8.46	6\7	214.7°*	7.4°*	13.1°*	27°*	62.9°*	-6.7°*
Lee Peak area	86°22.16'S	152°03.01'W	SGB2A	112/15W*	felsic Wyatt Fm.			6\7	214.7°*	-7.2°*	13.2°*	26.9°*	62.6°*	0.6°*
Lee Peak area	86°22'S	152°03'W	SGB2B		felsic Wyatt Fm.		3.57	7\7	6.4°*	24.6°*	6.5°*	88.3°*	214.4°*	-9.3°*
Lee Peak area	86°22'S	152°03'W	SGB2B	112/15W*	felsic Wyatt Fm.			7\7	3.7°*	39°*	6.4°*	88.6°*	211.6°*	-18.4°*
Mt. Farley area	86°31.12'S	153°44.88'W	SGB2D		Wyatt Fm.		1.75	7	unstable	-	-	-	-	-
Mt. Farley area	86°31'S	153°44.8'W	SGB2E		Wyatt Fm.		2.93	7	unstable	-	-	-	-	-
Mt. Wyatt	86°46'S	154°00'W	SGB2F		Wyatt Fm.		2.31	6\7	191.2°	-0.4°	7.2°	87.0	37.2°	-3.0°

Table 1. (continued)

	Site Coordinates			Tilt-			NRM x						Paleopole	
	Latitude, deg.	Longitude, deg.	Site	Correction	Lithology	Component	10-2 A/m	*n/N*	*D* , deg	*I* , deg	a95	*k*	Longitude, deg.	Latitude, deg.
					Southwestern Scott Glacier Region									
Mt. Verlautz	86°45'S	153°30'W	SGB2G		Wyatt Fm.		3.11	6\7	188.0°	4.9°	5.7°	141.4	34.6°	-5.7°
Mt. Farley area	86°34.58'S	152°49.00'W	SGB2H		Wyatt Fm.		4.46	7	unstable	-	-	-	-	-
			SGB2A&G		Wyatt Fm.	B		6	192.3°	24.0°	15.1°	20.5	44.8°	-15.9°
					Mt. Zanuck Region									
Mt. Gerdel	85°59'S	149°30'W	BLK3A		altered Zanuck granite		0.021	7	unstable	-	-	-	-	-
Mt. Andrews	85°59.95'S	149°17.45'W	BLK3B		mafic enclaves in BLK3C		0.418	7	unstable	-	-	-	-	-
Mt. Andrews	85°59.95'S	149°17.45'W	BLK3C		Zanuck granite		0.312	6	unstable	-	-	-	-	-
Zanuck Peak East	85°57'S	151°53'W	BLK3D		Zanuck gr. and enclaves	A	0.265	9\15	204.1°	4.4°	12.2°	18.9	53.4°	-5.9°
Zanuck Peak East	85°57'S	151°53'W	BLK3D		Zanuck gr. and enclaves	D	0.265	12\15	192.4°	40.3°	5.2°	72.0	42.0°	-26.9°
Outlook Peak	85°59'S	150°50'W	BLK3F		Zanuck gr. (mafic phase)		0.357	7	unstable	-	-	-	-	-
Outlook Peak	85°59'S	150°50'W	BLK3G		diorite		167	6\6	201.9°	3.1°	3.5°	359.2	51.1°	-5.3°
unnamed	86°05'S	149°30'W	BLK4A		Zanuck granite		14.4	6\6	198.3°	-6.6°	5.5°	148.4	48.3°	-0.5°

Site coordinates were determined from handheld Global Positioning System (GPS) Magellan receiver or place name geographic directory; site locations are shown in Figure 1. NRM, natural remanent magnetization; n/N, number of samples used for site mean calculation/total number of samples; D, site mean declination; I, site mean inclination; a95, semi - angle of the cone of 95% confidence about the site mean; k, estimate of precision parameter; VGP, virtual geomagnetic pole. Positive values are in the Northern Hemisphere and negative values are in the Southern Hemisphere.

* Data are discussed in the text.

Table 2. Scott Glacier Location Mean Directions and Paleopoles

	Approximate										Reliability Criteria							
Lithology	Age*, Ma	*n/N*	*D, deg*	*I,* deg	a95, deg	*k*	PPLon, deg	PPLat, deg	A95, deg	*R*	1	2	3	4	5	6	7	Q
Zanuck granite	513-516	9\54	183.8°	7.2°	10.8°	23.7	36.2°†	-7.3°†	8.6°†	8.7822	x	x	x			x	?	4
Wyatt and Ackerman Formations	525 ± 3	8\48	189.9°	10.1°	16.7°	12.0												
		8\48	190.9°‡	-7.7°‡	10.8°‡	27.1‡	40.3°	2.7°	9.1°	7.8170	x	x	x	x	?	x	x	6
Wyatt, Ackerman,	525 ± 3	11\69	188.6°	3.8°	13.1°	13.1												
and Mt. Paine/Johansen Peak		11\69	189.3°‡	-9.0°‡	7.9°‡	34.7‡	39.8°§†	1.3°§†	6.0°§†	10.8329	x	x	x	x	?	x	x	6

Abbreviations are as follows: *n/N* , number of sites used for calculation of mean direction and paleopole/total number of samples represented (the left-hand value is the number of sites used to calculate listed means); *D* , declination; *I* , inclination; a95, error circle at the 95% confidence level; *k* , estimate of precision parameter; PPLon and PPLat, paleopole longitude and latitude; A95, radius of error circle at 95% confidence level; R, resultant vector; PLat, paleolatitude. Reliability criteria (Q factor) are based on the work of Van der Voo [1988]: 1, well-determined age; 2, samples >24, k>10, A95<16°; 3, demagnetization published in sufficient detail; 4, positive field tests (fold, conglomerate, and baked contact); 5, tectonic coherence with block or craton; 6, presence of antipodal reversals; 7, lack of similarity with poles for younger times. The Zanuck pole is similar to the Early Ordovician Gondwana pole at the A95 level but not at the A63 level.

* The Wyatt (in a general sense) age represents the mean between 524 ± 2 and 526 ± 2 Ma.

† Values are final poles.

‡ Data are tilt-corrected from Ackerman Ridge.

§ These poles use the tilt-corrected virtual geomagnetic poles from Ackerman Ridge.

Peak is considered a primary layering formed owing to compaction of the ash-flow tuff [*Borg*, 1980].

Two components of magnetization were observed in some samples of the Wyatt Formation. The first, lower -temperature component (B) was found in a few samples and was south - directed and moderately shallow downward pointing. A high - temperature component (A) at Mount Wyatt, Mount Verlautz, and the Lee Peak ridge sites yielded south -directed orientations with very shallow, mostly upward pointing inclinations (Table 1).

The results from the Wyatt Formation containing felsic lenses on the Lee Peak ridge were less well - defined than the massive Wyatt Formation sampled at Mount Wyatt and near Mount Verlautz. These sites (SGB2A and SGB2B), without a tilt - correction, produced similar directions to those of the massive Wyatt Formation (Table 1). Site SGB2B was of opposite polarity (north and shallowly downward pointing) to the other Wyatt sites. Tilt- correction of the felsic layers causes greater scatter in the data. The shallow dip in the felsic layers is considered to be due to differential compaction (welding above 600°C [*Bierwirth*, 1982]) of the ash flow tuff along the flanks of a volcano. The Wyatt Formation at Ackerman Ridge, after tilt correction (based on the conformably overlying Ackerman Formation strata), yields a high- temperature component that is south-directed and upward pointing like the directions observed at Mount Wyatt and the non- tilt-corrected Wyatt at Lee Peak (Table 1).

The Wyatt Formation sites at Mount Mooney and Mount Farley did not yield meaningful directions. In the case of Mount Farley, sites SGB2E and SGB2F were magnetically unstable. The three sites sampled at Mount Mooney (Table 1) were in very coarse pyroclastic material and did not produce consistent within-site directions.

The mean unblocking temperature for the Wyatt Formation high - temperature "A" component was ~550°C whereas the mean unblocking temperature for the "B" component was ~350°C (Figure 2a). The age of magnetization of component A may be close to the emplacement age of 526 Ma obtained from the extrusive Wyatt Formation rocks at site LGC5A. An $^{40}Ar/^{39}Ar$ isotopic age on biotite from a roof pendant of Wyatt Formation within granite at Mount Griffith indicates a cooling age of 500 Ma [*Baldwin et al.*, 1995] (Figure 1) and is part of a younger overprinting assemblage. The component B directions may reflect overprinting from the younger, undeformed granites, emplaced primarily in the Cambro-Ordovician.

4.2. Ackerman Formation

Three lava flows were sampled from the Ackerman Formation along Ackerman Ridge and are described here on the basis of the stratigraphic section published by *Stump* [1983]. The first site, LGC5B, was the stratigraphically lowest volcanic unit (152 m thick) near the contact with the Wyatt Formation [*Stump*, 1983]. The second site, LGC5C, is from the next volcanic unit, mapped as 292 m thick [*Stump*, 1983]. The last site, LGC5D, was sampled from a volcanic unit that is 310 m thick [*Stump*, 1983].

These three sites yielded multicomponent directions. The high - temperature component, A, occurred between 500° and 570°C (Figure 2b). Using a bedding correction (see Table 1) based on the sedimentary strata between the flows, stable directions were found that were south- directed and shallowly upward - pointing. The age of magnetization for the component A, if primary, should be very close to the Pb-Pb zircon age of 524 ± 2 Ma from site LGC5C for these rapidly cooled extrusive rocks.

Component C, found in 19 samples, yielded very steep, upward - pointing inclinations without tilt - correction. The dispersion increases with tilt-correction (Table 1). The pole determined from the component C (without tilt- correction) is similar to those observed for the late Tertiary [*DiVenere et al.*, 1994] and may reflect some overprinting due to the emplacement of the Miocene mafic volcanic rocks found nearby (Figure 1).

4.3. Mount Paine

The Mount Paine (Figure 1) tonalite intrudes the Wyatt Formation although close intermixing of the tonalite with the Wyatt Formation led *Stump et al.* [1986] to infer that intrusion of the tonalite occurred while the Wyatt Formation was not entirely crystallized and hence represented the same magmatic episode. Although mapped as Mount Paine type granite, we found the Johansen Peak sites (LGC3C & LGC3D) to resemble intrusive Wyatt Formation. Four sites were collected in this area, and while very weak magnetically, three sites did yield stable, single- component, south- directed, shallowly- upward pointing directions (Table 1 and Figure 2c).

U-Pb zircon isotopic dating on the Mount Paine tonalite yielded a lower intercept age of ~531 Ma ±5/50 (not concordant) for site LGC3A [*Encarnación and Grunow*, 1996]. If the Mount Paine tonalite is comagmatic with the Wyatt Formation, its age would be better approximated by the 526 Ma Wyatt Formation age.

4.4. Zanuck Granite

The Zanuck granite was described by *Stump* [1995] as a gray, coarse, K-feldspar megacrystic (up to 10 cm long), cordierite, blue quartz-bearing hornblende, biotite granite. We believe that the Zanuck granite is exposed not only from Mount Griffiths to the Mount Zanuck area [*Stump*, 1995] but also between Price Bluff and Mount Noville (Figure 1) [*Encarnación and Grunow*, 1996]. U-Pb isotopic data from the Zanuck granite (site BLK3C) near Outlook Peak (Figure 1) yielded zircon crystallization ages of 521± 2 Ma (2 σ error based on two concordant fractions) [*Encarnación and Grunow*, 1996]. An $^{40}Ar/^{39}Ar$ biotite plateau age of 496 ± 3 Ma (2σ; nine steps representing >90% of ^{39}Ar; a standard equivalent to McClure Mountain hornblende MMhb-1 at 523 Ma was used) was obtained from site LGC1H at Mount Noville (Figure 1).

Eight sites were collected in the Zanuck granite in the Price Bluff to Mount Bowlin region (Figure 1). Of these eight sites, six yielded stable magnetic directions that were shallowly downward - pointing and south - directed (Table 1 and Figure 2d). One of these sites, LGC1E, was of opposite polarity to the other granite sites. The remaining two sites were too weak for measurement. In the Mount Zanuck area, seven sites were collected but only three sites yielded stable magnetic directions. One of these sites, BLK3G, was in a small, megacrystic diorite intruding the Zanuck granite. On the basis of field relationships, we believe the diorite is a younger, more mafic phase in the Zanuck granite.

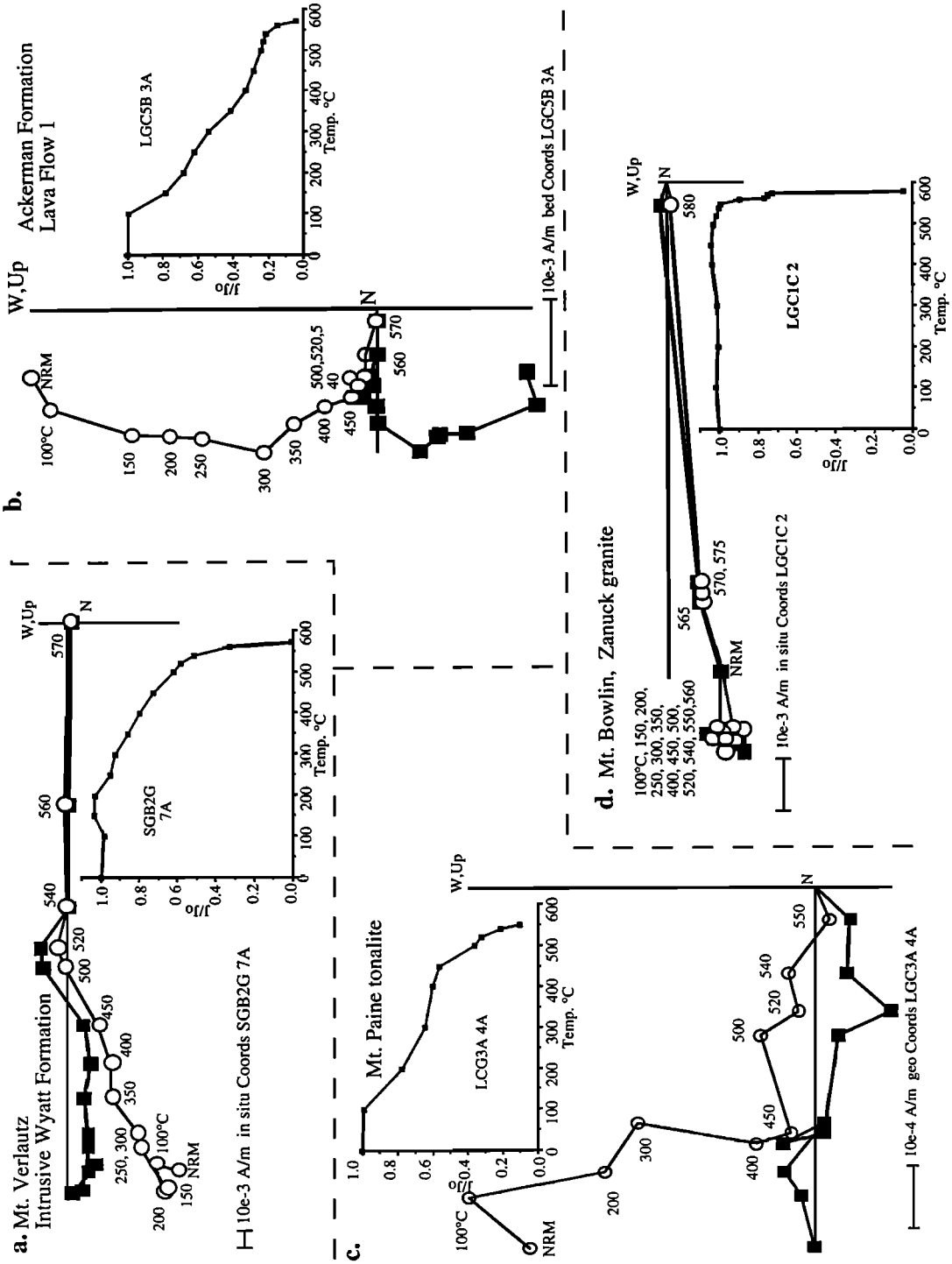


Figure 2. Orthogonal projection of vector endpoint diagrams showing demagnetization behavior of samples from the Scott Glacier area: (a) Intrusive Wyatt Formation; (b) Ackerman Formation lava flow; (c) Mount Paine tonalite; and (d) Zanuck granite. Open circles (solid squares) are projections on the vertical (horizontal) planes at the indicated levels of thermal (°C) demagnetization. Normalized intensity (J/Jo) plots are shown for lithologies where thermal demagnetization was used.NRM, natural remanent magnetization.

The maximum unblocking temperature for the nine magnetically stable granitoid sites falls between 540° and 580°C. If uniform cooling is assumed for the Zanuck granite, emplaced at 650°C at 521 ± 2 Ma (based on the pressure - temperature (P-T) data mentioned in the regional geology section) and cooling to 300°C (argon closure temperature in biotite) by 496 ± 3 Ma, a cooling rate of ~14°C/Myr is obtained. This cooling rate estimate produces a magnetization age between 513 and 516 Ma using unblocking temperatures between 540° and 580°C. We consider this to be a minimum age for the magnetization because cooling was probably faster initially, rather than uniform.

5. Summary of New Paleomagnetic Data From the Scott Glacier

Similar paleomagnetic directions were obtained from 11 sites in the Mount Paine tonalite, Wyatt and Ackerman Formations. The paleomagnetic sites from the comagmatic Wyatt and Ackermann Formations pass a fold test indicating that component A predates the folding event (Table 2). The folding occurred prior to the emplacement of the undeformed, equigranular Early Ordovician granites [*Borg*, 1980]. In addition to the positive fold test, the Wyatt Formation contained one site in the felsic layered volcanics near Lee Peak that was of opposite polarity. Given the positive fold test, the presence of reversals, the high unblocking temperature, and the different lithologies sampled, we believe that component A in the Wyatt and Ackerman Formations plus the single -component magnetization recorded in the Mount Paine/Jacobsen Peak intrusive rocks represents a primary magnetization. The 11 sites from these locations were averaged to obtain an average "Wyatt" (in a general sense) mean direction and "Wyatt" mean pole (determined from the site mean VGPs) (Table 2 and Figure 3).

The high-precision U-Pb ages obtained on these rocks (526±2 and 524±2 Ma on the Wyatt and Ackerman Formations, respectively [*Encarnación and Grunow*, 1996]) indicate that they are late Tommotian to early Atdabanian in age using the *Bowring and Erwin* [1998] timescale. Given the extrusive nature of the Ackerman and most of the Wyatt sites, it seems likely that the age of magnetization of the primary directions listed in Table 2 are the same as the U-Pb ages. The isotopic age of the Mount Paine tonalite is less well -constrained (lower

Table 3. Gondwana Paleomagnetic Poles in an African Reference Frame

Country	Formation/Unit	Age*		PPLong, deg	PPLat, deg	A95, deg	Q	References
			Middle Ordovician (478-458 Ma)					
Australia	Stairway Sandstone	Middle Ordovician		34.0°	47.0°	3.0°		*Embleton* [1972]
			Early Ordovician (490-478)					
		Mean (~485 Ma)	*N* =4	6.0°	34.0°	8.0°		*Grunow* [1999]
			Late Cambrian-Early Ordovician (~490 Ma)					
Australia	Chatswood Limestone	Camb-Ordovician		41.0°	51.0°	6.0°		*Ripperdan and Kirschvink* [1992]
Antarctica	Wyatt Fm., B comp.	~500 Ma?		32.0°	38.0°	-		this paper
			Middle Cambrian (510-500 Ma)					
Australia	Hugh River	Middle Cambrian		13.0°	62.0°	9.0°	3	*Embleton* [1972] and *Lindsay* [1987]
			Late Early Cambrian (526-510 Ma)					
Africa	Ntonya ??†	522 ± 13 Ma		345.0°	28.0°	2.0°	6	*Briden et al.* [1993]
India	Jutana	late Early Cambrian		11.0°	3.0°	12.0°	4	*Klootwijk et al.* [1986]
India	Salt Pseudomorph	late Early Cambrian		354.0°	-3.0°	5.0°	4	*Wensink* [1972] and *Jell and Hughes* [1997]
India	Purple Sandstone	late Early Cambrian		352.0°	4.0°	11.0°	3	*McElhinny* [1970]
Australia	Kangaroo Isand†	late Early Cambrian		356.0°	14.0°	8.0°	3	*Klootwijk* [1980]
Australia	Billy Creek†	late Early Cambrian		1.0°	11.0°	14.0°	6	*Klootwijk* [1980]
Antarctica	Sor Rondane†	~515 Ma		8.0°	11.0°	5.0°	3	*Zijderveld* [1968]; and *Grunow* [1995]
Antarctica	Sor Rondane†	~525 Ma		15.0°	6.0°	8.0°	5	*Rolf and Henjes-Kunst* [1998]
Antarctica	Zanuck granite	~515 Ma		17.0°	43.0°	9.0°	4	this paper
Antarctica	Wyatt/Ackerman Fm.	525 ± 3 Ma		14.0°	52.0°	6.0°	6	this paper
		Mean (~518 Ma)	*N* =4	5.0°	11.0°	10.0°		this paper, excludes Ntonya pole
		Mean (~518 Ma)	*N* =5	1.0°	14.0°	13.0°		this paper, with Ntonya pole
			Early Early Cambrian (543-526 Ma)					
Africa	Ntonya ??	522 ± 13 Ma		345.0°	28.0°	2.0°	6	*Briden et al.* [1993]
Australia	Hawker Group (base)	early Early Cambrian		351.0°	26.0°	11.0°	5	*Klootwijk* [1980]
Australia	Antrim Plateau	early Early Cambrian		314.0°	17 0°	13.0°	5	*McElhinny and Luck* [1970]
Australia	Upper Arumbera	early Early Cambrian		338.0°	-12.0°	4.0°	6	*Kirschvink* [1978] *and Lindsay* [1987]
Australia	Todd River	early Tommotian (~530)		337.0°	-8.0°	7.0°	7	*Kirschvink* [1978]

PPlong. and PPlat. are the paleopole longitude and latitude, respectively. Only poles with (Q) quality ratings of 3 or higher are listed (see Table 2 for criteria). Q ratings for poles published prior to1992 are from *Van der Voo* [1993]. The following are the rotation poles used to bring all poles into an African reference frame: *Royer and Sandwell* [1989], India to Antarctica (10.9°E, 7.8°N, -65.1°); and Australia to *Antarctica (38.9°E, -2.0°S, -31.5°); Lawver and Scotese* [1987], Antarctica to Africa (328.58°E, -7.78°S, 58.0°).

*Numeric ages for the poles are based on the *Bowring and Erwin* [1998] timescale.

†Poles are not used to calculate the Gondwana mean poles.

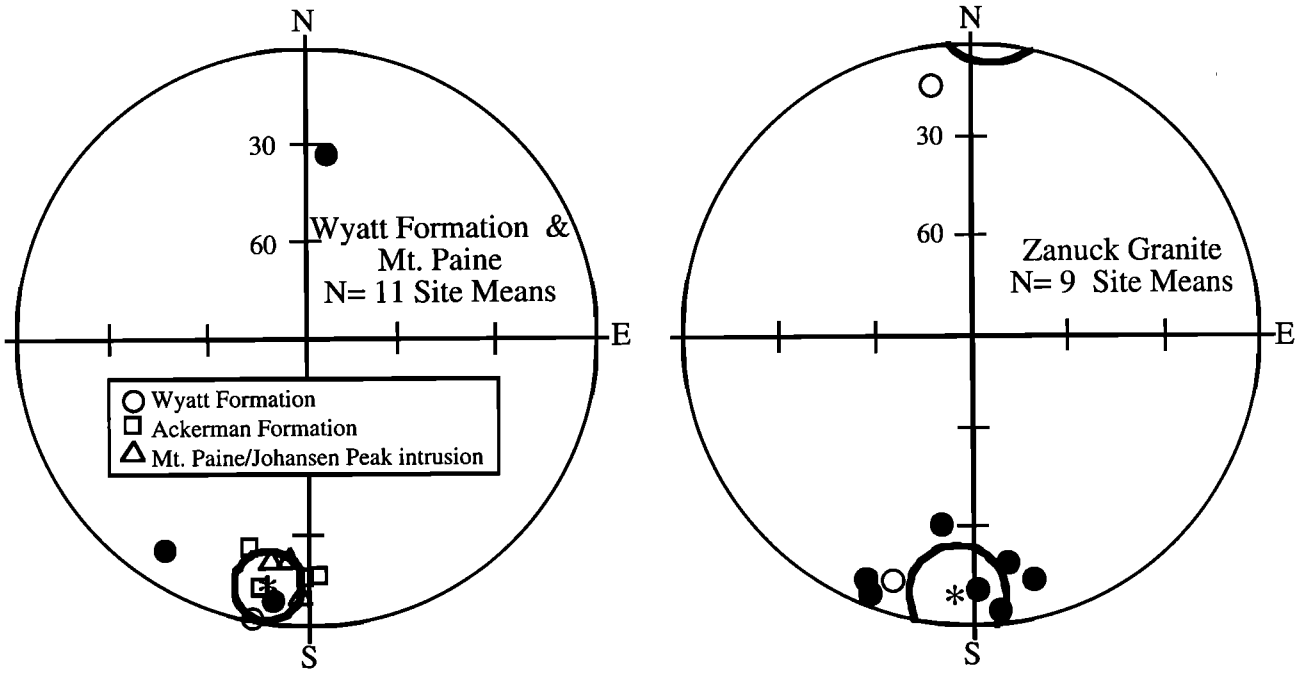


Figure 3. Equal angle projections of site mean directions from the Wyatt Formation, Ackerman Formation, Mount Paine tonalite, Johansen Peak intrusive Wyatt Formation, and the Zanuck granite. Open (closed) symbols are on the upper (lower) hemisphere. Circle represents the a95 radius of 95% confidence for the directional means.

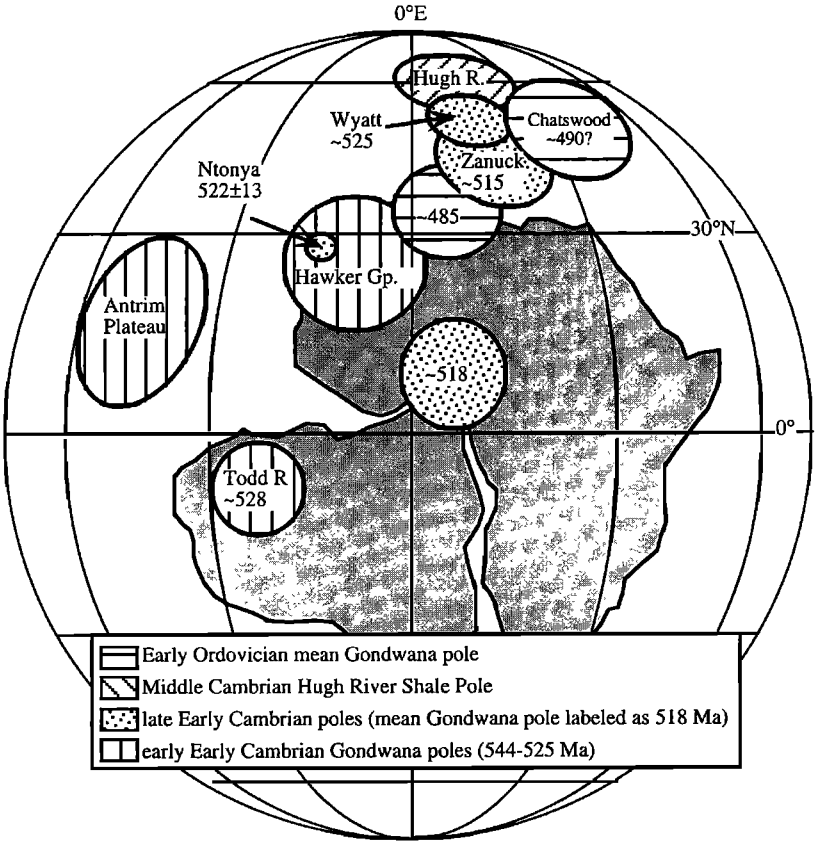


Figure 4. Comparison of Gondwana mean Cambrian and Early Ordovician poles with the Wyatt and Zanuck poles (in an African reference frame; see Table 3 for references and rotation poles). The Wyatt and Zanuck poles are most similar to the Hugh River Shale (HRS) pole [*Embleton*, 1972] rather than the Gondwana means. The Todd pole is the Todd River late Early Cambrian pole used by *Kirschvink et al.* [1997] to support the true polar wander model (Table 3).

intercept ~531±5/50 Ma), but on the basis of field relationships, the tonalite is comagmatic with the Wyatt Formation and hence, approximately the same age.

The similarity of the component B pole from the Wyatt Formation with the Late Cambrian Chatswood Limestone pole [*Ripperdan and Kirschvink*, 1992] and the Middle Ordovician Australian Stairway Sandstone pole [*Embleton*, 1972] (Table 3) may indicate Late Cambrian to Ordovician magnetic overprinting related to the Ross and Delamerian Orogenies in Antarctica and Australia.

The Zanuck granite sites yielded consistent between- site directions in nine sites (Figure 3). While there are no paleohorizontal indicators in these rocks, the flat-lying Devono-Carboniferous Beacon Supergroup strata overlying granite at Mount Blackburn indicate that there has been little or no tilting since the mid-Paleozoic over much of the Zanuck granite field area. Tilting may have occurred prior to Beacon strata deposition, but the consistency of the data over such a large region and across major glacial divides (Figure 1) argues against that scenario. We feel that the single- component, high unblocking temperature, dual- polarity magnetization found in the relatively unaltered Zanuck granite is reasonable evidence that the Zanuck granite has not been magnetically overprinted. While the Zanuck pole overlaps the Early Ordovician mean Gondwana pole at the A95 level (Table 3 and Figure 4) it does not overlap it at the A63 level, hence we believe that the Zanuck pole is not similar to the Early Ordovician pole and has not been remagnetized.

6. Comparison With Gondwana Cambrian Poles

Using the Cambrian *Bowring and Erwin* [1998] timescale, a new Gondwana mean pole has been calculated for the late Early Cambrian (~525-510 Ma, Atdabanian and Botomian) based on reference poles from Antarctica and Australia (Table 3 and Figure 4). Gondwana poles from the early Early Cambrian (543-525 Ma) are not in particularly good agreement with each other, and hence no mean has been calculated. The early Early Cambrian poles from the Amadeus Basin agree with each other [*Kirschvink*, 1978], but they do not agree with the Australian Hawker Group or the African Ntonya pole (assuming an older age) [*Klootwijk*, 1980; *Briden et al.*, 1993] (Table 3). Poles from the Salt Ranges in Pakistan [*McElhinny*, 1970; *Wensink*, 1972] have been omitted from the means because of documented younger Tertiary rotations [*Klootwijk et al.*, 1986]. The new ~525 and ~515 Ma paleopoles from the Scott Glacier region are different from the late Early Cambrian paleopoles from the Sør Rondane Mountains, Antarctica, and also from any well - constrained Early Cambrian poles from Gondwana (Table 3).

7. Discussion

Assuming that the Wyatt and Zanuck poles represent primary magnetizations that have not been structurally disturbed, there are several models that can explain our results. The Wyatt and Zanuck poles may indicate that the Scott Glacier area was a terrane relative to East Gondwana during the Early Cambrian. Another possibility is that the mean poles used for the Cambrian Gondwana apparent polar wander path have missed rapid plate motions caused by either normal tectonic mechanisms or inertial interchange true polar wander [*Kirschvink et al.*, 1997]. Lastly, it is possible that there has been vertical axis rotation of the region if the Wyatt pole is compared with the African Ntonya pole [*Briden et al.*, 1993] rather than with the other temporally equivalent poles from Australia and Antarctica.

7.1. Terrane Accretion and the Ross Orogeny

The new 525 Ma Wyatt pole may indicate that the Scott Glacier area is an allochthonous terrane relative to cratonic East Antarctica. The idea of late neo-proterozoic to Cambrian terrane accretion in the Transantarctic Mountains is not a new one. On the basis of isotopic data, *Borg et al.* [1990] suggested that a Beardmore microcontinent along the TAM was accreted to Antarctica by middle Early Cambrian time. The trend of the isotopic signatures include the Scott Glacier in the Beardmore microcontinent [*Borg and DePaolo*, 1994]. *Borg and DePaolo* [1991] also suggested that north Victoria Land, Tasmania, and eastern Australia may all have been part of a collage of terranes with the Beardmore microcontinent.

Rowell and Rees [1989] also proposed that some of the volcanics-dominated units were part of a terrane accreted to cratonic East Antarctica by the late Middle Cambrian. They emphasized the difference in facies between an inboard marginal cratonic series and an outboard volcanic/sedimentary sequence (termed the Queen Maud Terrane) [*Rowell et al.*, 1992]. Recently, *Duebendorfer and Rees* [1998] suggested that the Ellsworth-Whitmore Mountains (EWM) terrane in West Antarctica (Figure 1) was geologically linked to the Queen Maud Terrane rather than to East Antarctica. *Goodge et al.* [1993] found structural evidence for transcurrent movement of deep-seated metamorphic rocks along the Miller Range shear zone (Figure 1) in the central TAM, which they related to Early Cambrian oblique plate convergence along the East Antarctic margin during the Ross Orogeny. *Goodge et al.*, suggest that rocks outboard of the Miller Range shear zone (Figure 1) may have been allochthonous and were accreted to the Antarctic margin in a sinistral transpressional regime between ~540 and ~520 Ma. Hence several authors using different lines of evidence have suggested the possibility that some of the rocks in the Transantarctic Mountains may be allochthonous.

Our new data can be interpreted to support the terrane accretion concept. Since our new data from the Scott Glacier area come from within the Queen Maud Terrane of *Rowell et al.* [1992], we also call the hypothesized allochthonous block the Queen Maud Terrane (QMT). Given the similarity in geologic histories of the Queen Maud terrane with parts of Tasmania and Australia [*Borg and DePaolo*, 1991] and the uncertainty in the location (existence?) of parts of West Antarctica, we have speculatively shown them as being part of a greater Queen Maud Terrane (which in itself is a collage of terranes). The EWM terrane is shown in two possible locations, one linked to Gondwana [*Dalziel and Grunow*, 1992] and the other linked to the Queen Maud Terrane [*Duebendorfer and Rees*, 1998].

In the models shown in Figure 5, latest neo-proterozoic metasedimentary rocks found in the Transantarctic Mountains (i.e., the Goldie Formation, possibly equivalent to the Scott Glacier La Gorce Formation) are thought to have been deposited along a rifted East Antarctic passive margin opposite to

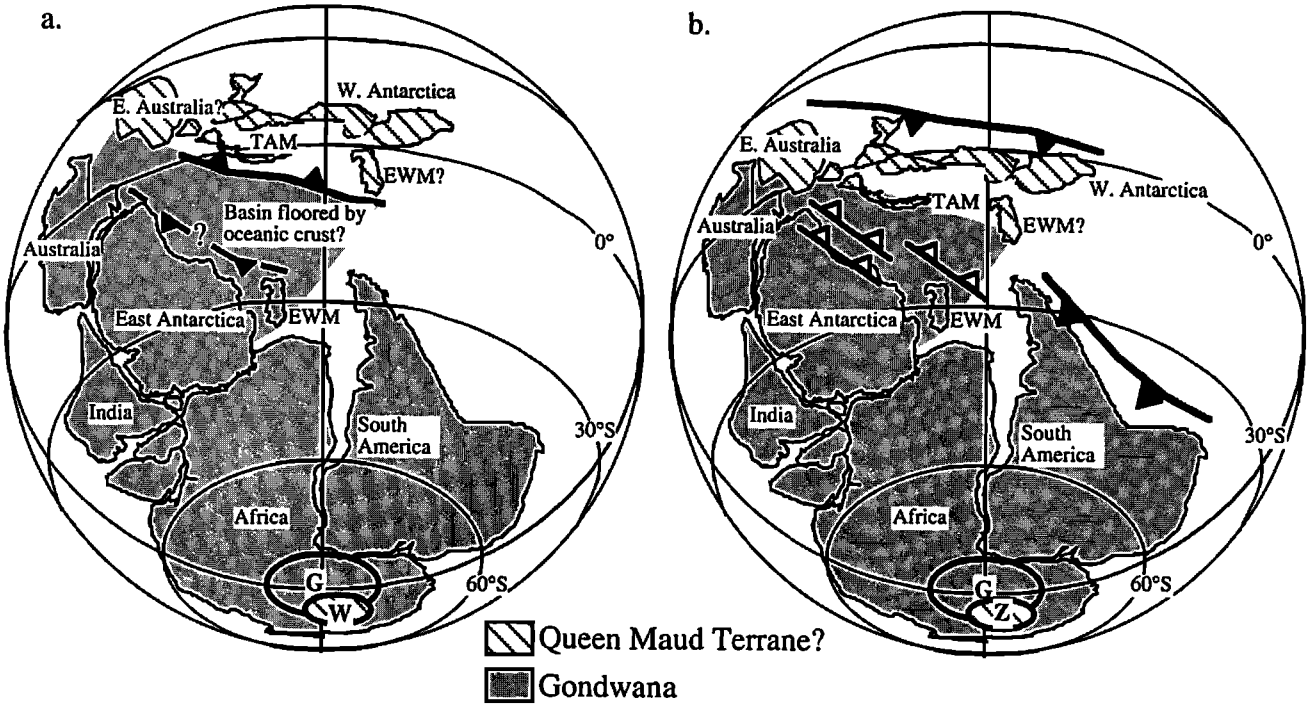


Figure 5. Plate reconstruction of Gondwana using the Wyatt and Zanuck poles, respectively, and centered on the Gondwana late Early Cambrian reference pole (Table 3). It is assumed that a basin existed between East Gondwana and the Queen Maud Terrane (QMT) in the late neo-proterozoic and Early Cambrian. The reconstructions have been made to minimize the width of the basin while maintaining overlap between the Gondwana (G) and Wyatt (W) or Zanuck (Z) poles. The two locations shown for the Ellsworth - Whitmore Mountains terrane (EWM) are based on the suggestion of *Duebendorfer and Rees* [1998] that it was part of the QMT and on the suggestion of*Watts and Bramall* [1981] that it was adjacent to East Antarctica. Closure of the basin and accretion of the QMT to East Gondwana caused the deformation known as the Ross Orogeny. TAM, Transantarctic Mountains. (a) At ~525 Ma a significant oceanic basin may have existed between the QMT and East Gondwana. Closure of this basin may have occurred by northward subduction beneath the QMT causing the Wyatt magmatism (and perhaps by southward subduction beneath East Antarctica?). (b) At ~515 Ma the basin was mostly closed between the QMT and East Gondwana. Subduction may have jumped outboard after accretion of the QMT, contributing to later magmatism.

Laurentia as proposed in the Southwest U.S. - East Antarctica (SWEAT) model [*Dalziel*, 1991; *Hoffman*, 1991; *Goodge*, 1997]. We suggest that the conjugate margin to East Antarctica in the latest neo-proterozoic may have been the QMT rather than Laurentia. The crustal nature (oceanic or continental) of any basin between the QMT and East Antarctica is unknown.

In Figure 5a, we show a marginal basin floored by oceanic crust. It is conceivable that this basin was floored by thinned continental crust, but given the width of the basin suggested by the paleomagnetic data, it would be difficult to close the basin without subduction. The Wyatt and Zanuck igneous activity in the QMT may have been generated by oblique northward subduction beneath the QMT and possibly southward subduction? beneath the Transantarctic Mountains (Figure 5a). In Figure 5b the basin has narrowed, and the QMT has "docked" with East Antarctica. Subduction may then have jumped outboard of the QMT, thus contributing to Middle Cambrian magmatism identified elsewhere in the Transantarctic Mountains [e.g., *Encarnacion et al.*, 1999].

One problem with the terrane hypothesis is the location of the suture between the QMT and East Antarctica. If the terrane concept is correct, the suture must lie inboard (south) of the Scott Glacier (Figure 1), but its projection along the Transantarctic Mountains is uncertain. The suture could follow the isotopic boundaries through the Miller Range [*Borg and DePaolo,* 1991] or it could swing outboard near/beneath the Ross Ice Shelf/Ross Sea in the south Victoria Land (SVL) and central TAM sectors or it could be located cratonward of the TAM beneath the East Antarctic ice sheet. Each of these options would have major implications for the regional geology of the TAM.

If the QMT did exist, it was presumably accreted sometime during the Ross Orogeny, which ended by ~500 Ma [*Encarnación and Grunow*, 1996]. After the Early Ordovician, magmatism either ceased or moved outboard of the Queen Maud terrane into West Antarctica [*Grunow et al.*, 1996].

7.2. True Versus Apparent Polar Wander

Another explanation for the discrepancy between the Scott Glacier and Gondwana Cambrian data sets is that the Gondwana mean poles are flawed because they average too much time, are based on poles that are not well - constrained in age, or have magnetic overprints. If the current Early Cambrian Gondwana

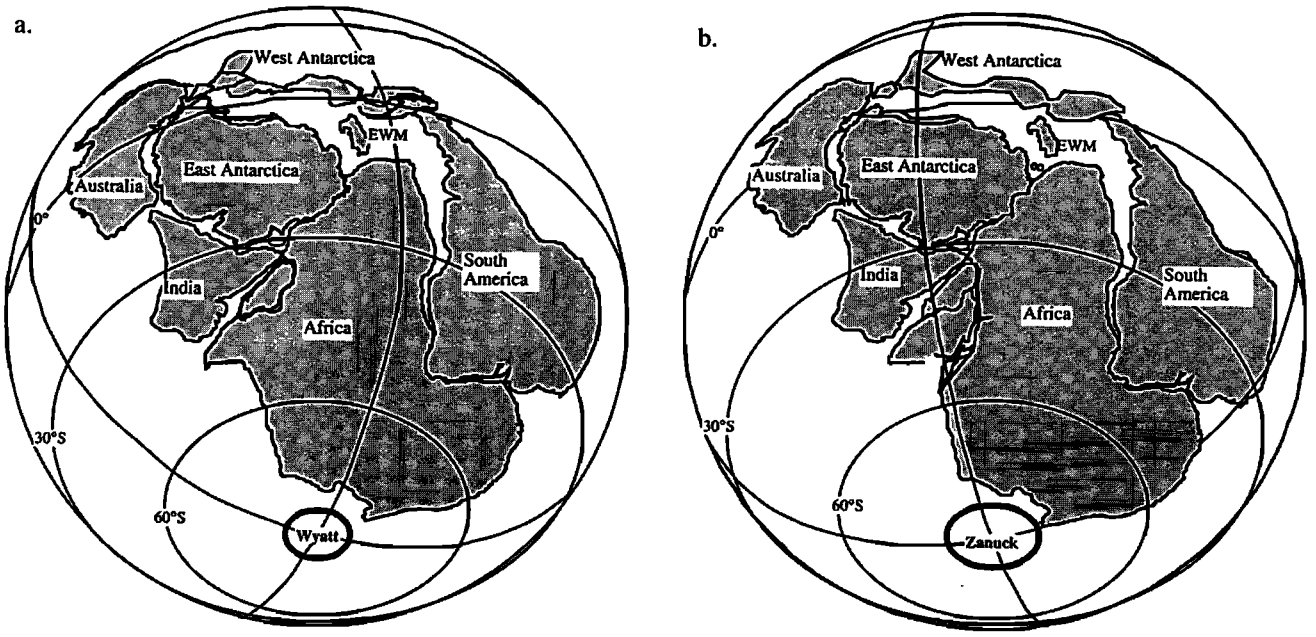


Figure 6. Reconstructions showing the location of Gondwana assuming the Wyatt and Zanuck poles are representative of Gondwana reference poles: (a) ~525 Ma and (b) 515 Ma. Note that southern West Gondwana would have been in significantly more equatorial latitudes in these reconstructions versus the ones in Figure 5 using the Gondwana mean for 518 Ma.

mean poles are flawed or miss a rapid polar wander interval for some reason, then perhaps the Wyatt and Zanuck poles do record a period of rapid apparent polar wander in the late Early Cambrian. In this scenario the Scott Glacier area does not need to be a terrane, and the Wyatt and Zanuck poles may be more representative of Gondwana during the 525 - 515 Ma period than the Australian and Antarctic late Early Cambrian Gondwana poles listed in Table 3. Reconstructions using the 525 Ma Wyatt and ~515 Zanuck poles as Gondwana reference poles are shown in Figures 6a and 6b. The more equatorial position of the southern Africa and southern South America areas for the ~525 and ~515 Ma interval differs from other Early Cambrian reconstructions [*Scotese and McKerrow*, 1990].

Another possibility is that the late Vendian to Cambrian Gondwana apparent polar wander path (APWP) (along with paleopole data from other non-Gondwana continents) indicates that the Earth's lithosphere and mantle rotated 90° between ~530 and ~515 Ma, leading to inertial interchange true polar wander (TPW) [*Kirschvink et al.*, 1997]. The Gondwana evidence for the TPW hypothesis comes exclusively from Australia and relies primarily on two poles: the latest Tommotian (530-525 Ma) Todd River dolomite [*Kirschvink*, 1978] and the Late Cambrian-Early Ordovician (~490 Ma) Chatswood Limestone pole [*Ripperdan and Kirschvink*, 1992]. Almost all of the other Early Cambrian Gondwana poles listed in Table 3 that encompass the proposed TPW interval do not appear to show evidence for the TPW event. However, these other Gondwana poles may not be representative of the late Early Cambrian as noted by *Kirschvink et al.* [1997], leaving open the question as to whether there is evidence from Gondwana for a TPW event. The new 525 Ma Wyatt and ~515 Ma Zanuck poles fall in the proposed time period of TPW. Using the Todd River, Wyatt, and Zanuck poles (Figure 4) as reference poles for 528, 525, and 515 Ma, respectively, requires rapid latitudinal shifts for East Gondwana and a large rotation, perhaps consistent with the TPW model. Clearly, the TPW idea requires additional testing from other Gondwana continents and non-Gondwana continents as well.

7.3. Vertical Axis Rotations

Vertical axis rotation of the Scott Glacier area can improve the agreement of the Wyatt pole with the temporally equivalent Todd River pole or the Ntonya pole but not the Gondwana mean late Early Cambrian pole listed in Table 3 and shown in Figure 4. Comparison of the Wyatt pole (~525 ± 3 Ma) with the African Ntonya pole (522 ± 13 Ma) could allow a 32° vertical axis rotation of the Scott Glacier area relative to East Antarctica. Restoring the Wyatt pole to the Todd River pole (~528 Ma) requires a ~110° rotation, creating anomalous structural trends and isotopic boundaries in the region [*Borg and DePaolo*, 1991], hence we do not consider this a geologically realistic option.

The significant differences in rotation sense and amount between the supposedly similar age Ntonya and Todd River poles create uncertainty about the magnetization ages of these poles. If the Ntonya pole is truly late Early Cambrian (~522 Ma) and has not experienced structural disturbance, then the supposedly temporally equivalent late Early Cambrian Australian and Antarctic poles (Table 3) indicate either rapid polar wander during this interval or problems in the assigned magnetization ages. A simpler answer to the discrepancy between the Ntonya pole and almost all of the other Gondwana poles may be found in assigning an older age to the Ntonya pole (using the isotopic

errors). The Ntonya pole would then be virtually identical to the early Early Cambrian Australian Hawker Group pole. The Ntonya rocks are thought to have been emplaced at great depth and cooled very slowly, making it difficult to know the age of magnetization accurately. Any vertical axis rotation, if it occurred, would have probably ceased by the end of the Ross Orogeny in the Late Cambrian [*Encarnación and Grunow*, 1996] as there are no other major tectonic events after this time that affected the entire Scott Glacier area.

8. Conclusions

The Wyatt and Zanuck paleomagnetic data yield high - quality Cambrian poles that differ from the Gondwana Cambrian reference poles. We believe it is unlikely that these poles represent remagnetization events given the positive fold test, presence of reversals, and lack of similarity to younger poles. There are several scenarios that can explain the discrepancy between the new Antarctic poles and the Gondwana Cambrian paleopoles. The first possibility is that they indicate that the Scott Glacier area and, by geologic correlation, other areas of Antarctica and Australia? are part of the Queen Maud Terrane that was allochthonous to East Gondwana in the Early Cambrian. Convergence and collision of the QMT provide one tectonic explanation for the Cambrian Ross Orogeny in Antarctica. If, however, some of the Gondwana reference poles are flawed and the Wyatt and Zanuck data represent East Antarctica, then three possibilities arise. First, the new data may indicate rapid apparent polar wander (without true polar wander) in the Cambrian. This would place parts of West Gondwana in more equatorial latitudes in the late Early Cambrian. Second, the new poles could be interpreted to broadly support the true polar wander hypothesis [*Kirschvink et al.*, 1997]. Third, it is possible that the area experienced a vertical axis rotation if the Wyatt pole is compared with the African Ntonya pole [*Briden et al.*, 1993]. However, the Ntonya pole is not similar to any other late Early Cambrian pole from Gondwana. Without additional Cambrian paleomagnetic data from Gondwana and geologic data from the Transantarctic Mountains, it is difficult to choose between the terrane motion, polar wander, and vertical axis rotation options at this time.

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